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Introduction

The first sitting of this specification for Pearson Edexcel International GCSE Geography took place in 2019. This Examiner's report is intended to provide an insight into performance on Paper 1 Physical Geography, in particular, analysing the majority of questions in terms of what went well and where common mistakes and underperformance were evident.

This paper consists of two sections from which candidates answer two 25 mark questions from Section A and one 20 mark question from Section B, the total marks on the paper are 70.

The exam includes multiple-choice questions, short, open response, calculations and extended response questions. The exam command words which are used in the paper are defined in the specification. Each of the questions is mapped to one or more of the Assessment Objectives (AOs).

In section A river environments, coastal environments and hazardous environments are covered. Candidates are required to select two out of three questions.

In section B candidates choose one out of three fieldwork related questions relating to river environments, coastal environments and hazardous environments.

It is important that candidates focus on the requirements for each command word and the Assessment Objective (AO) distribution, particularly for the longer response 8-mark questions, to ensure they access the full range of marks available.

Question 1(b)ii

Candidates accessed this question well and were able to state one characteristic of the channel shape in the lower course of a river. On occasion some candidates identified a characteristic of the river valley rather than the channel shape.

Question 1(c)

A significant proportion of candidates were able to achieve 2 marks on this question, where they were required to explain one reason why some sediment is transported by traction. A minority of responses focused on erosion processes.

(c) Explain **one** reason why some sediment is transported by traction in rivers.

(2)

Due to hydraulic action, it works by pushing the waters in the river floor, which sediments can be created by this factor.

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0 marks This response has explained the incorrect process.

(c) Explain **one** reason why some sediment is transported by traction in rivers.

(2)

Sediments can be transported through traction due to erosional processes like attrition and abrasion, in which ~~the~~ the water flow will transport sediments throughout the river course.

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0 marks. Incorrect process explained.

(c) Explain **one** reason why some sediment is transported by traction in rivers.

(2)

Because it is too heavy and cannot floar so it gets dragged along the river bed due to the current.

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2 marks. It is too heavy (1) and so it gets dragged along the river bed (1).

(c) Explain **one** reason why some sediment is transported by traction in rivers.

(2)

As river flows, there are sediments from river bed and banks. It gets moved over the river by traction

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1 mark. Sediments get rolled over the river (1).

Question 1(d)

This question required candidates to use a resource (Figure 1a) showing a stacked bar graph for current and future water demand. The first mark was awarded for identification of a relevant piece of evidence from the resource, in this case identifying a category which had increased between the two years shown on the graph and a second mark for a reason why this increase may have occurred.

A large proportion of candidates were able to achieve 4 marks, where this was not the case it was often as they only identified a change and did not develop this to give a reason for the change. A very small proportion of candidates wrote a response focusing on irrigation which was the only category to decrease between the two years shown, and the question focused on reasons for increase in water demand not a reason for a change in water demand and so a decrease in demand could not be awarded credit.

(d) Study Figure 1a in the Resource Booklet.

Suggest **two** reasons why demand for water is projected to increase.

You **must** refer to the resource in your answer.

(4)

1 Because electricity need water , manufacturing of products also ~~require~~ ~~require~~ ~~to~~ require's water

2 Water is required for 'mostly everything e.g drinking , washing bathing ~~cleaning~~ cleaning et.

ResultsPlus
Examiner Comments

0 marks. There is no reference to the resource in either point and the statements made are not specifically linked to why demand has increased as uses for water are stated.

(d) Study Figure 1a in the Resource Booklet.

Suggest **two** reasons why demand for water is projected to increase.

You **must** refer to the resource in your answer.

(4)

1 Electricity, in figure 1a it shows a large increase in water demand for electricity as urbanisation requires electricity

2 manufacturing, to make manufacturing more sustainable it is expected to increase in figure 1a

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2 marks. Large increase in water demand for electricity (1). Manufacturing is expected to increase (1).

(d) Study Figure 1a in the Resource Booklet.

Suggest **two** reasons why demand for water is projected to increase.

You **must** refer to the resource in your answer.

(4)

1 Increase in population. If there is more people, more water is going to be demanded by individuals

2 More urbanisation. More houses are going to be built, and more electricity is going to be demanded for those houses, and that requires more water.

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3 marks. Increase in population (1) - the development simply repeats that there are more people rather than linking this factor to evidence from the resource which is required for the second mark. More electricity (1) due to an increase in the number of houses being built (1).

(d) Study Figure 1a in the Resource Booklet.

Suggest **two** reasons why demand for water is projected to increase.

You **must** refer to the resource in your answer.

(4)

1 Increased demand for domestic usage as people are able to afford more water due to higher income as countries develop, so they use more water in their home not having to worry too much to save water. Household sizes will increase, so more people use more water, increasing demand

2 Increased demand in livestock as population of the world increases, They demand more food so more livestock is needed, increasing demand.

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4 marks. Increased demand for domestic use (1) as people can afford more water due to higher income (1). Increased demand for livestock (1) as population increases (1).

Question 1(e)

The majority of candidates achieved the mark for this question, a few candidates identified the landform as a meander which was not correct for an upland river landscape shown.

Question 1(f)

This question required candidates to explain how one climate and one slope factor can have an effect on river processes. Generally, candidates coped with the demand of this question well. A proportion of candidates did not gain full marks as their development was not linked to river processes instead focusing on the impact on discharge which was not enough to show they understood the impact on river processes.

(f) Explain how **one climate** factor and **one slope (gradient)** factor can affect river processes.

(4)

climate

one Climate factor can affect the river process by having harsh winds this means that, there will be a change in the river flow which will affect the river processes

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0 marks. Harsh winds is too vague to credit as a climate factor affecting river processes. This would be more applicable to coasts/wave processes.

(f) Explain how **one climate** factor and **one slope (gradient)** factor can affect river processes.

(4)

climate

Heavy rainfall will increase the discharge of the river.

↳river flows faster / velocity increases

↳river may flood

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Examiner Comments

1 mark. Heavy rainfall (1) - the development is linked to drainage basin processes/flood risk/river regime rather than the influence on river processes.

(f) Explain how **one climate** factor and **one slope (gradient)** factor can affect river processes.

(4)

climate

Heavy / Prolonged Rainfall will cause the infiltration rate to exceed the drainage capacity, which can lead to flooding as the drainage densities get exceeded by water. extra.

slope (gradient)

Steep slopes can allow for quicker water flow to go into the drainage basins and nearby tributaries, increasing the risk of flood as

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2 marks. Heavy rainfall (1) - the development is linked to drainage basin processes/flood risk and not linked to the impact heavy rain can have on river processes. Steep slopes (1) - again the development is not linked to the impact on river processes and so cannot be awarded credit.

(f) Explain how **one climate** factor and **one slope (gradient)** factor can affect river processes.

(4)

climate

Hot & dry climates can lead to increase in evaporation (rate) of water in the channel network, decreasing river discharge ~~level - less press~~ (& hence river velocity), leading to less hydraulic action & abrasion (erosion).

slope (gradient)

A deeper gradient / steeper slope means that river velocity increases, allowing for faster transport of eroded material as well as increased hydraulic action in the upper course.

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Examiner Comments

4 marks. Hot/dry climates (1) decreases discharge, decreasing hydraulic action (1). Steeper slope (1) increases velocity, increasing transport (1).

(f) Explain how **one climate** factor and **one slope (gradient)** factor can affect river processes.

(4)

climate

Increased rainfall leads to increased river discharge, and this may cause more hydraulic action or other fluvial processes such as erosion.

slope (gradient)

Slopes encourage mass movement (soil creep in particular), and this causes weathered material to slope down under the influence of gravity. This contributes to increased river load.

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Examiner Comments

4 marks. Increased rainfall (1) increases discharge which increases hydraulic action (1). Mass movement (1) as material slopes down under the influence of gravity (1) - subaerial processes are included in river processes and this response answers the question in a different style to most candidates but has a clear understanding of how a slope influences the process of mass movement.

Question 1(g)

For this question candidates were required to explain how one source of river pollution can affect water quality. It is often these 3-mark questions which students find most demanding as they often find it challenging developing one point in sufficient depth to be awarded 3-marks. However, there was more evidence of candidates being awarded 3 marks showing an improvement in their depth of understanding.

(g) Explain how **one** source of river pollution can affect water quality.

(3)

People may throw rubbish in the ~~into~~ river ~~and~~.

ResultsPlus
Examiner Comments

1 mark. People may throw rubbish into the river (1) - this is just enough to be awarded a mark for a specific type of pollution.

(g) Explain how **one** source of river pollution can affect water quality.

(3)

One source could be industrial waste as it pollutes the water therefore it leads to aquatic wildlife dying. Industrial waste is waste that comes from businesses such as manufacturing.

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Examiner Comments

2 marks. Industrial waste (1) leads to aquatic wildlife dying (1).

(g) Explain how **one** source of river pollution can affect water quality.

(3)

Agriculture can lead to leachy when ~~pesticides and~~ fertilisers moves into rivers after a flood. This increases algeae growth which leads to eutrophication, causy decrease in oxygen levels in water for ~~a~~ animels.

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Examiner Comments

3 marks. Agriculture (1) causing leaching of fertilisers into water (1) which increases algae growth/eutrophication (1).

Question 1(h)

These 8-mark extended writing questions candidates are required to blend their use of the resource (AO4) with their own knowledge and understanding of the issue presented (AO3), therefore they are not case study questions, rather they require the candidates to apply their geographical understanding to the context shown in the resource.

Many candidates clearly used the resource to describe the strengths of the flood prevention method shown, using the textboxes most often to discuss the pros and cons of the strategy. Some candidates also demonstrated their understanding of the hydrograph shown to support their points.

To access the higher marks candidates needed to go beyond simply describing what is shown in the resources but provide several clear developed reasons for the differences in the river regimes shown. The command word 'analyse' needs to be addressed to achieve full marks and often candidates found this demanding as there was often minimal judgement demonstrated in candidate responses.

(h) Study Figure 1c in the Resource Booklet.

Analyse the strengths of this flood prevention strategy.

You **must** refer to the resource in your answer.

(8)

To use Aswan High Dam. Regulated the flow of water and provided a year-round supply for ~~irrig~~ ~~irrig~~ irrigation.

[Screenshot of candidate work] | *images/4GE1_01R_Q01h_1182701_0561_000964509_006.png*

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Examiner Comments

0 marks. This sentence is just a direct lift from the resource and therefore is not awarded credit.

(h) Study Figure 1c in the Resource Booklet.

Analyse the strengths of this flood prevention strategy.

You **must** refer to the resource in your answer.

(8)

In figure 1c the Aswan high Dam was built to lower and controll the nile river which has been doing worth somes maxtum discharge at 14000 but it can also drop to as low as 27m which is almost concidder as a drought. the Aswan Dam was 1 billion dollars to build and since then it has saved lives since 1965, 1975 and 1987, less flooding reduces cost for flood repairs and insurance. On the graphe it shows that since the Dam was built it has kept the discarge as even as possible which benefits alot from having a water source to something protecting us. the Dam was built from Alexandria from mediterarian sea all the way to lake Nasser.

[Screenshot of candidate work] | *images/4GE1_01R_Q01h_2222865_0561_001957929_006.png*

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Examiner Comments

Level 1 - 2 marks. There are some small changes to the wording in the first 8 lines of the response to demonstrate more than direct lift from the resource and the last 4 lines describing the hydrograph and location demonstrate more engagement with the resource. There is no real AO3 in this response, but it is awarded the middle of level 2 for its engagement with the resource.

(h) Study Figure 1c in the Resource Booklet.

Analyse the strengths of this flood prevention strategy.

You **must** refer to the resource in your answer.

(8)

~~comparatively , the~~ Before the Aswan high dam was built, the River Nile flooded nearby places frequently ~~which~~ and unstably, ~~it~~ with discharge as high as 14 000 m^3 / second . But ~~comparatively~~ comparatively , after the dam was built , ~~it~~ it regulates the flow of water ~~it~~ stably and provide farmers with efficient amount of water for irrigation . ~~This~~ ~~changes~~ ~~these~~ From these changes , we could see that the dam is quite effective on controlling the discharge of the river .

From the river regime , we could see that the average discharge after the dam was built is way more stable than the average discharge before the dam was built , this also shows ~~it~~ the effectiveness of the dam .

However , the dam ~~causes~~ costs too long to be built , in a period of 10 years . This may ~~cause~~ ~~cause~~ cost too long ~~and~~ for people to enjoy the benefits .

But overall , the dam is effective on it's ~~imp~~ impacts brought after built . ~~It~~ ~~not~~ Not only did it ensure the safety of people , but also the life of them .

[Screenshot of candidate work] | *images/4GE1_01R_Q01h_1120690_0561_000834457_006.png*

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Level 2 - 4 marks. This response has a balance of AO3 and AO4 throughout the answer. There are three ideas supported with evidence from the resource and evidence of considering the dams effectiveness. The depth of AO3 is too weak for each idea to move higher into level 2.

(h) Study Figure 1c in the Resource Booklet.

Analyse the strengths of this flood prevention strategy.

You **must** refer to the resource in your answer.

(8)

Figure 1c shows the observations before and after Aswan High Dam was built, providing additional maps and graphs.

Figure 1c ~~shows~~ graph shows a significant decrease in average discharge, particularly from August to November. This shows the dam was able to ~~store~~ regulate the amount of water supplied compared to amount demanded. Dam acts as a barrier to enter water so they are still able to have water in case of drought.

Figure 1c also stated that there is less flooding, meaning less repair and insurance costs. This means government expenditure is lower, so they are able to instead spend on other problems.

However, it could be argued that it's ~~not effective~~ too expensive (cost 1 billion USD to build). The government could spend on other ways to treat flooding.

Figure 1c also stated that it took 10 years to complete meaning that during time taken, they would need a lot of spending to treat floods and drought, increasing overall lag time.

Overall, Figure 1c also stated that there were low water removal from Jan to August. This heightens the risk of drought, however, with

the use of the dam water could be regulated, meaning they could allow in more water that was stored in supply. This also decreases risk of unemployment for farmer as it allows irrigation to occur throughout the year.

(Total for Question 1 = 25 marks)

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Level 2 - 5 marks. This response has a range of ideas all linked to evidence provided in the resource. There is some judgement demonstrated in the strengths of the dam being built with a judgement for each point made. This response needs more depth of explanation for each point made to move to the top of level 2.

(h) Study Figure 1c in the Resource Booklet.

Analyse the strengths of this flood prevention strategy.

You **must** refer to the resource in your answer.

(8)

Dams can be very effective for flood prevention, as they can store water for irrigation and ~~th~~ control the river flow.

Firstly, according to Figure 1c, the Aswan High Dam ~~locates in the upper or course of the river where there~~ can help regulate the flow of water. Dams can help to block the water when there is rapid river ~~flood~~ flow. Therefore, it can slow down the river regime when there is rapid surface runoff. As a result, it avoids to much discharge in the river and prevent from flood. From Figure 1c, ~~what~~ we can observe that the average discharge before Aswan High Dams were built ~~was~~ a range between around 70 - 700m^3 per day, while the average discharge after building has a range ~~be~~ around 100 - 250 m^3 per day. This shows that the discharge can be remained in a low level without any sudden floods.

Secondly, the Aswan High Dam can help to avoid any damage caused by flooding. Heavy floods can easily flood low-lying ~~are~~ areas and damage peoples houses. Therefore, people ~~are~~ become homeless and may rely on ~~temporay~~ temporary shelthers, and aid. Therefore, there will be a high cost to provide aids and repair the infrastructures, such

as housing and even destroy agriculture and industrial production. ~~This dam helps to provide~~ Since the Aswan High Dam helps to regulate the water flow, the risk of flooding can be reduced and prevent damaging of instructure. This protects people property and

(Total for Question 1 = 25 marks)

Egypt can prevent from economic loss.

To conclude, the Aswan Pam can help to regulate the river flow which can ~~avoid~~ reduce risk of flooding. Therefore, the infrastructure in Egypt will not be easily damage avoiding economic loss. It also contains ~~a~~ other benefits, such as saving water supply for irrigation, ~~a~~ meaning that they can continue to plant crops for food and reduce death risks.

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Level 2 - 6 marks. This response has two fairly well developed ideas linked to the change in discharge and reduction of damage as strengths of the dam. There is evidence from the resource provided to support each point explained. There is a concluding statement at the end of the response demonstrating enough judgement to move the answer to the top of level 2. To reach level 3 there needs to be a wider range of ideas discussed.

(h) Study Figure 1c in the Resource Booklet.

Analyse the strengths of this flood prevention strategy.

You **must** refer to the resource in your answer.

(8)

~~Flood~~ The strengths of flood prevention strategies depend on many factors. They depend on the time taken to complete, the effectiveness of the scheme, ~~and the~~ impacts of floods, and the cost needed.

The flood prevention scheme had many benefits in Egypt which was the construction of the Aswan Dam. Before the dam was completed, there were many ~~problem~~ issues. The River Nile flooding was shown to not be controlled effectively prior to the construction of the Dam as the discharge could be as high as 14000 m^3/second in Fig 1c. This caused many devastating floods which affected agriculture from farmers, industries and housing as shown in Fig 1c. Crops were destroyed which may have reduced the income of farmers while industries were disrupted which could have led to job losses. Homelessness would have been another impact as it was shown in Fig 1c that housing was affected. This could have made costs of repair very expensive for the Egyptian government. However, there were also periods of intense drought during low water periods as the discharge could fall as low as 279 m^3/second. Water supply wasn't properly regulated so the supply of water may not be enough for Egypt's population. Therefore, more people will suffer from dehydration while farmers lose income due to bad harvest.

However, since the dam was completed, Egypt was able to properly regulate the flow of water. It was shown in Fig 1c that the dam provides a year-round supply for irrigation, this meant that farmers can grow crops around the year **(Total for Question 1 = 25 marks)**

more easily because they will have an adequate water supply. The dam was also very effective as it ~~managed~~ managed to save lives in many time periods in 1965, 1975 and 1987. The highlights ~~how effective~~ the high quality and effectiveness of the Aswan High Dam scheme. Since Dams ~~are~~ are hard engineering structures that hold back an area of flood water, they can stop it from reaching certain areas ~~the~~ to. This makes the costs ~~to~~ of repair lower for the government and it puts less pressure on insurance companies.

However, in fig 1c, the dam cost around $1 billion USD which puts pressure on taxpayers. The dam also took 10 years to complete as shown in fig 1c which means that ~~the people~~ flooding the effects of flooding would not be reduced over a long time.

Overall the ~~effects of the~~ benefits of the scheme ~~outweigh~~ outweigh the negatives as many benefits were observed after the construction of the dam.

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Level 3 - 8 marks. This response is structured in a different way to most where the problems of not having a dam are discussed in depth and this is then linked to the reasons why building the dam was a good thing to do. There is evidence used in both halves of the response to support their ideas and then a more substantiated conclusion reaching a judgement over whether the strengths outweigh the negatives. This was just enough to move the response to the top of level 3.

Question 2(a)ii

In this question candidates were required to state one other impact of sea level change on coastal landforms. Most candidates were awarded the mark for this question with the most common responses being raised beaches. A few candidates misinterpreted the question as asking for a cause for sea level rise and were not awarded credit. A few other candidates stated an impact on the human environment and not coastal landforms as requested in the question.

Question 2(c)

In this question candidates were required to explain one type of soft engineering used in coastal management. Most candidates were awarded at least 1 mark as they were able to name a soft engineering strategy, however, a significant proportion of candidates only went on to describe the strategy rather than explain how it works to reduce erosion/flooding (and therefore manage the coast).

(c) Explain **one** type of soft engineering used in coastal management.

(2)

creating sea walls to protect the coast line from floods and tsunamis

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0 marks. This is not relevant to a question on soft engineering.

(c) Explain **one** type of soft engineering used in coastal management.

(2)

Beach replenishment when sand is manually pumped onto existing beaches

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1 mark. Beach replenishment (1)

(c) Explain **one** type of soft engineering used in coastal management.

(2)

beach replenishment: pumping sand onto a beach, so it can absorb waves energy and reduce rate of erosion.

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2 marks. Beach replenishment (1) so it can absorb waves energy (1).

Question 2(d)

This question required candidates to use a resource, Figure 2a, showing two photographs of different hard engineering strategies. The first mark was awarded for identifying a feature shown on the resource that could cause conflict, and the second mark was awarded for the development of this point. Candidates generally engaged with the resource well and identified a feature which would be likely to cause conflict. Many candidates identified specific key groups who would be at conflict (which was not required for this question to gain credit, as no key groups were shown/implied from the resource) and demonstrated their secure understanding of conflict in coastal environments.

Where candidates did not gain full marks this was most often due to not having identified a conflict based on evidence shown in the resource.

(d) Study Figure 2a in the Resource Booklet.

Suggest **two** reasons the coastal management shown could cause conflict with different users of the coastline.

You **must** refer to the resource in your answer.

(4)

1 Firstly since the management has caused a disturbance in the landscape of the beach, which could mean that they can attract (less) tourist causing (holiday) business to loose out on ~~opportu~~ opportunities to generate sales.

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1 mark. Disturbs the landscape so less tourists attracted (1) - no reference to the resource provided.

(d) Study Figure 2a in the Resource Booklet.

Suggest **two** reasons the coastal management shown could cause conflict with different users of the coastline.

You **must** refer to the resource in your answer.

(4)

1 One reason it could cause conflict with different users is due to aesthetic reasons. Many people believe that coastal management looks untidy and ruins the coastline, causing conflict.

2 Another reason it could cause conflict is because it can destroy local animal habitats. This could cause conflict between users as they do not agree with disturbing the wildlife.

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2 marks. Due to aesthetic reasons as people believe it looks untidy (1) - this response does not have clear evidence from the resource. Can destroy habitats (1) - again without providing any evidence from the resource.

(d) Study Figure 2a in the Resource Booklet.

Suggest **two** reasons the coastal management shown could cause conflict with different users of the coastline.

You **must** refer to the resource in your answer.

(4)

1 In fig 2a it shows a ~~grow~~ groine and a sea wall. This can ~~cause~~ prevent longshore drift and erosion of the coastline at that point, but the problem will move down the coastline instead, which can be a conflict

2 Another conflic is that the beach is not attractive. This means that it won't attract tourists and people there ~~won't~~ can't do business

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3 marks. Groyne (1) can reduce sediment supply downstream (1). Not attractive (1) - with no clear evidence from the resource.

(d) Study Figure 2a in the Resource Booklet.

Suggest **two** reasons the coastal management shown could cause conflict with different users of the coastline.

You **must** refer to the resource in your answer.

(4)

1 one reason is the groynes are unattractive this may upset the resident living in the houses facing the coast as unappealing and visually polluting.

2 Another reason might be the seawall beach is unaccessable for the tourist and people as there isn't convience for them to go to the coast of the beach.

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4 marks. Groynes (1) are visually polluting (1). Sea wall (1) makes beach inaccessible (1).

Question 2(e)

For this question candidates were required to explain how one abiotic characteristic can affect a named coastal ecosystem. It is often these 3-mark questions which students find most demanding as they often find it challenging developing one point in sufficient depth to be awarded 3-marks. Historically, candidates have found explaining the role of abiotic factors more challenging therefore, it was pleasing to see that candidates generally coped with the demands of this question well. There was almost no evidence of candidates identifying a biotic factor in their initial point and this meant many candidates were awarded at least 2 marks. Some did find it a challenge to create the chain of consequence required for these 3-mark questions.

(e) The non-living parts of an ecosystem are called abiotic characteristics.

Explain how **one** abiotic characteristic can affect a named coastal ecosystem.

(3)

Named coastal ecosystem

Coral reefs

Pollution as an abiotic characteristic effects the coral reefs by causing potential coral bleaching. Resulting in a less diverge coral reef ecosystem.

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0 marks. This response does not contain a specific abiotic factor and so was not awarded credit.

(e) The non-living parts of an ecosystem are called abiotic characteristics.

Explain how **one** abiotic characteristic can affect a named coastal ecosystem.

(3)

Named coastal ecosystem

Mangroves

mangroves need at optimum temperature of around 19° C which would provide them with good optimal growth atmospheres and a perfect salinity of sea water to help them grow

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1 mark. Need an optimum temperature of 19 degrees (1).

(e) The non-living parts of an ecosystem are called abiotic characteristics.

Explain how **one** abiotic characteristic can affect a named coastal ecosystem.

(3)

Named coastal ecosystem

sunlight

Coral reefs require large amount of light, so they only survive in the ~~area~~ shallow sea with 0.5 to 2.5 metres height to the sea level, and the range could be slightly above or under due to the tidal.

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2 marks. Large amount of light (1) and so live in shallow water (1).

(e) The non-living parts of an ecosystem are called abiotic characteristics.

Explain how **one** abiotic characteristic can affect a named coastal ecosystem.

(3)

Named coastal ecosystem

Coral reefs

Temperature can affect coral reefs, as they must in waters from 23-29°C, ~~to~~ and if the temperature is too high, the zooxanthellae in the coral polyps will be expelled, and the coral will be bleached, causing inhabitants of the reef to ~~need to~~ be expelled.

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3 marks. They thrive in waters 23-29 degrees (1) if temperature is too high Zooxanthellae is expelled (1) the coral will be bleached (1).

Question 2(f)

The majority of students correctly identified the labelled country with the shortest length of saltmarsh ecosystem.

Question 2(g)

This item required candidates to explain two threats caused by tourism to a coastal ecosystem. Generally, this question was answered well by candidates and although 'such as coral reefs' was stated in the question responses about any coastal ecosystem were credited. Most candidates coped with the demands of this question well, although, it would be helpful to remind candidates pollution needs to be qualified to be awarded credit. Some candidates found it challenging to develop their initial point, for example some stated a tourist threat but did not go on to explain the impact of this threat on the ecosystem. Other candidates stated the impact of the threat eg reduces biodiversity but did not identify a specific tourism threat.

(g) Explain **two** threats caused by tourism on coastal ecosystems such as coral reefs.

(4)

1 Coral reefs usually grows under the sea

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Examiner Comments

0 marks. No relevant response linked to a threat by tourists.

(g) Explain **two** threats caused by tourism on coastal ecosystems such as coral reefs.

(4)

1 Pollution - tourist's will come to see coral Reefs Themselfs and when they do sometimes they don't clean up ~~all~~ after themselfs polluting the Area.

2 Damaging and Breaking coral reefs, tourists come And damage or Break the coral Reefs to take home as a suvanier.

ResultsPlus
Examiner Comments

1 mark. Pollution on its own is too vague for credit as a threat and so the first part of the response was not awarded any credit. Tourists damage/break the coral (1)

(g) Explain **two** threats caused by tourism on coastal ecosystems such as coral reefs.

(4)

1 pollution from the suncreams that the tourists use when they explore the coral reefs.

2 Tourists may take pieces of the coral with them as a souvenir, this can damage the coral

ResultsPlus
Examiner Comments

2 marks. Pollution from suncream (1) Pieces are taken as a souvenir (1)

(g) Explain **two** threats caused by tourism on coastal ecosystems such as coral reefs.

(4)

1 When divers or snorkelers view the coral, thy can accidentaly kick the coral with their flippers which can kill the coral

2 Higher demand for hotels and infastructre requires more land so people may need to clear coastal ecosystems such as mangroves by deforestation.

ResultsPlus Examiner Comments

3 marks. Divers kick coral (1) causing it to die (1). People clear land for hotels (1) - responses focused on two different coastal ecosystems were awarded as the question stem states 'such as..' which is a steer for the candidates but does not mean they can only write about the ecosystem stated.

(g) Explain **two** threats caused by tourism on coastal ecosystems such as coral reefs.

(4)

1 Tourist can touch the corals, which will cause them to bleach, ~~otherwise~~ in other words die.

2 Tourists can throw the rubbish into the sea, which will pollute the coral reef habitat. Toxic chemicals released can also bleach corals.

ResultsPlus
Examiner Comments

4 marks. Tourists touch corals (1) which cause them to die (1). Throw rubbish introduces toxins (1) causing corals to bleach (1).

Question 2(h)

In these 8-mark extended writing questions candidates are required to blend their use of the resource (AO4) with their own knowledge and understanding of the issue presented (AO3), therefore they are not case study questions, rather they require the candidates to apply their geographical understanding to the context shown in the resource.

In this instance candidates were required to study Figure 2c, a topographical map, photograph and fact file of a coastal location in Canada.

Some candidates used the information from the resource and made links to the relevant physical processes responsible for the landforms shown along the stretch of coastline. Candidates' understanding of physical processes, particularly transport/deposition/erosion is strong and many were able to link these to the information provided in the resource. A number of candidates were also able to accurately discuss the role of the correct, sub-aerial processes involved in the formation of the coastal landscape shown.

The command word 'analyse' needs to be addressed to achieve full marks, a proportion of candidates demonstrated making a simplistic judgement but fewer demonstrated the analyses required to reach Level 3.

(h) Study Figure 2c in the Resource Booklet.

Analyse the effect of physical processes on the coastline.

You **must** refer to the resource in your answer.

(8)

There is a fetch km a
mean of 6km meaning that it
is sheltered and there is less
longshore drift

[Screenshot of candidate work] | *images/4GE1_01R_Q02h_402331_0561_000435734_011.png*

ResultsPlus
Examiner Comments

Level 1 - 1 mark. A basic idea linked to evidence from the resource.

(h) Study Figure 2c in the Resource Booklet.

Analyse the effect of physical processes on the coastline.

You **must** refer to the resource in your answer.

(8)

A spit is formed at the Gulf Island National Park Reserve. This is because of the process of longshore drift through the head land as we can see in Figure 2c. Not only a large spit is formed a ~~lagoo~~ Lagoon is made too. these types of Coastline characteristics are mainly ~~and~~ made through constructive waves.

[Screenshot of candidate work] | *images/4GE1_01R_Q02h_1133981_0561_000831127_011.png*

ResultsPlus
Examiner Comments

Level 1 - 2 marks. A basic response with evidence from the resource and a correct process involved in the spit formation to move the response to the middle of level 1.

(h) Study Figure 2c in the Resource Booklet.

Analyse the effect of physical processes on the coastline.

You **must** refer to the resource in your answer.

(8)

The large hook spit in the north is formed by ~~a~~ the longshore drift across the shore which carries sediments along the share by constructive waves. ~~It~~ Constructive waves has strong swash and a weaker backwash. ~~which~~ This forms a hook like spit which is shown in the figure 2c. This spit could protect ~~waves~~ strong waves reaching the National Park on the land, ~~damaging crops~~ which would protect vegetations on the land. It could ~~as~~ also protect from ~~deff~~ deforestation as the land is not that high ~~above~~ above Sea level.

ResultsPlus
Examiner Comments

Level 1 - 3 marks. This response was stronger on AO3 demonstrating knowledge of longshore drift and the characteristics of constructive waves, using evidence from the resource to reach the top of level 1.

(h) Study Figure 2c in the Resource Booklet.

Analyse the effect of physical processes on the coastline.

You **must** refer to the resource in your answer.

(8)

Physical processes on the coastline like spits and bars could be seen in the figure.

In figure 2c, A spit has formed on the coastline due to the transportation of sediment such as long shore drift, where waves hit the bay shore at prevailing wind direction and sediment builds up during the swash, sediment is carried straight back down again by backwash, this continues as sediment is transported in a zig-zag shape along the coast. The spit has a hooked end due to the wind. We can also see that the waves are constructive which means sediment is being built up on the coast.

In Figure 2c, we can see that the island is mainly made up of sand and clay which shows that the coast is eroded easier as the material is less resistant.

Sand dunes affect the coast as it causes friction with the water and acts as a natural sea defence. There is also vegetation growing, which may be biological

ResultsPlus
Examiner Comments

Level 2 - 4 marks. The first paragraph linked to longshore drift shows clear understanding of the process involved in creating the spit shown in the resource. With a second idea linked to the geology and influence of erosion. The second point is explained with less depth than the first point but there is evidence from the resource providing a balance of AO3 and AO4 to reach the bottom of level 2.

(h) Study Figure 2c in the Resource Booklet.

Analyse the effect of physical processes on the coastline.

You **must** refer to the resource in your answer.

(8)

figure 2c suggest that the dominant wave type of ~~a~~ the Sidney Island, Canada coast line is constructive wave. This suggest that more beach will be form on the coastline due to more deposition taking place as constructive wave have a strong swash and weak backwash. figure 2C also shown that a ~~spit has~~ long spit are found to the north of the island. this suggest that the ~~prevailing~~ longshore drift has occured this is ~~has~~ due to prevailling wind has caused sediment to be carry on to the beach at an angle and gravity has result the sediments to be carry down the beach at

carrying sediments in a zigzag pattern
a 90° angle. This process repeats along the coastline.
A spit is form when there is a sudden change in the coastline resulting in deposition of sediments as long shore drift still continues. figure 2c also suggest that an airport was built on the island suggest that the land must be flat. moreover, figure 2c suggest that the annual temperature is lower between 5-18°C therefore corals & mangrove wont be able to survive an physical process such a free thaw tende.

ResultsPlus
Examiner Comments

Level 2 - 5 marks. This response has one well developed idea linking the role of longshore drift to the coastal landscape shown and the evidence from the resource is interleaved into the response to clearly show understanding of the resource. They make a simple second point linking temperature to why certain ecosystems would not exist in this area. This is just sufficient to move to the middle of level 2.

(h) Study Figure 2c in the Resource Booklet.

Analyse the effect of physical processes on the coastline.

You **must** refer to the resource in your answer.

(8)

The term physical process refers to natural process that shape / influence the coastline. The figure above shows a island in Canada with a spit, a lagoon formed. As per the figure this coastline is primarily made of sand and clay, a type of softrock that is easily succesiptible to erosion, however the dominant wave type here is constructive waves meaning more deposition will happen and erosion will be lower. The sand beaches and sand dunes in this coastline will help stabilize

the coastline and reduce impacts of erosion. Towards the northern part of the island, a spit has been formed due to a change in the direction of the coastline leading to longshore drift depositing material there. Changes in wind speed and direction have caused a hook to be formed there. As a spit is created, it shelters an area behind the spit providing optimum conditions for a formation of a salt marsh. This is seen with the vegetation behind the spit. The salt marsh will reclaim more land for this island by trapping silt and mud through its creeks. it also provides a resting ground for many animals and fish which could be a reason why the area has been declared as a National park reserve.

In conclusion, this coastlines is primarily witnessing more deposition and depositional landforms, which will increase protection against natural phenomenons such as cyclones **(Total for Question 2 = 25 marks)**

The natural reserve can be used for tourism

ResultsPlus
Examiner Comments

Level 2 - 6 marks. This response shows a clear understanding of the role of different marine processes in the formation of the landscape shown whilst using evidence from the resource to support their explanation. There is some analysis language within the response and an attempt at a conclusion which isn't entirely linked to the focus of the question but is enough to just move this to the top of level 2.

(h) Study Figure 2c in the Resource Booklet.

Analyse the effect of physical processes on the coastline.

You **must** refer to the resource in your answer.

(8)

One effect is that the spit could become a bar which would trap salt water behind it forming a lagoon. The spit shown in Figure 2c is formed due to material by longshore drift deposition, and is curved inwards due to wave refraction. Over time, this process could continue, leading to the spit joining land, forming a bar. This could protect the Gulf Islands National park reserve shown in Figure 2c as it provides a barrier between the sea and the park reserve. Therefore if there was a tsunami, the bar might break the energy of it slightly and the impacts would be less severe.

Another effect is that Sidney Island might grow in size over the next hundred years. This is because the dominant wave type according to Figure 2c is constructive waves. This means that the swash is greater than backwash, and more material will be deposited instead of

eroded. The mean fetch is also quite small at 6km, this means that the strength of waves is not very strong and it is difficult to erode the material. However, Sidney Island is primarily made up of sands and clays according to Figure 2c. Clay is a soft rock and therefore erodes quite fast. This means that even if the dominant wave type is constructive, there might still be destructive waves. Therefore, the soft rock (clay) might still be eroded during that time, causing Sidney Island to not grow as big as it was supposed to. Another factor is that Sidney Island is quite low-lying, with its highest point being around 50 to 100 metres above sea level. This means that if sea levels rise due to melting ice caps caused by global warming, Sidney Island is at risk of coastal flooding. This might reduce their land size as more land is submerged.

However, it has a lot of vegetation according to Figure 2c, such as fir forests and ~~ovens~~ of oak woodlands. This might help prevent soil erosion as the roots of the vegetation are holding the soil together. This means less soil might be eroded. The vegetation could also help break the energy of the waves to prevent them ~~from~~ from eroding the material as much.

In conclusion, physical processes might help build onto/ save Sidney Island however factors such as sea level rise could cause coastal flooding and soil erosion.

ResultsPlus
Examiner Comments

Level 3 - 8 marks. The first paragraph shows clear understanding of the formation process of a spit with accurate processes identified. The second paragraph looks to the future and the role of the dominant wave type in the creation of the landscape. There is an explanation of the role of the fetch and geology in the landscape and a link to the role of the vegetation slowing erosion processes. This response has a more logical conclusion linked to the future factors outlined in the response which is sufficient to reach the top of level 3.

Question 3(b)ii

In this question candidates were required to state one hazard caused by an earthquake. Most candidates were awarded credit, however, some candidates confused impact with hazard. For these types of questions the physical hazard from the earthquake is required rather than buildings destroyed (which is an impact resulting from the ground shaking). The mark was awarded for the hazard responsible for the building destruction eg ground shaking/seismic waves.

Question 3(b)iii

Candidates were required to explain one long-term impact earthquakes can have on the natural environment. Most candidates were awarded marks for this question, however, a few explained an impact on people/human environment which was not awarded credit. Although, the term long-term is subjective in these questions it is important candidates are able to explain an impact which lasts for a prolonged period of time.

(iii) Explain **one** long-term impact earthquakes can have on the natural environment.

(2)

In forests an earthquake could destroy the trees which effects the life in the forest effecting the natrual enviroment

ResultsPlus
Examiner Comments

1 mark. Earthquakes could destroy trees (1).

(iii) Explain **one** long-term impact earthquakes can have on the natural environment.

(2)

Damaged Infrastructure :- Earthquake can damage the building this might take long ~~for the~~ to repair the buildings once again and can cost a huge amount.

ResultsPlus
Examiner Comments

0 marks. The focus of this question was impacts on the natural environment and this response explains an impact on the human environment and so was not awarded credit.

(iii) Explain **one** long-term impact earthquakes can have on the natural environment.

(2)

Earthquake can cause habitat destruction reducing biodiversity leading to extinction.

ResultsPlus
Examiner Comments

2 marks. Habitat destruction (1) reducing biodiversity/leading to extinction (1).

Question 3(c)

This item required candidates to use Figure 3a, which is a resource showing an information leaflet provided to residents in Salt Lake City. Candidates generally coped well with this question and were able to engage with the resource accurately. Some candidates only provided direct lifts in their responses and these are not awarded credit, it would be helpful to remind candidates that although they need to use evidence from the resource for these 4-mark questions there needs to be evidence of them using the information rather than just copying the text.

(c) Study Figure 3a in the Resource Booklet.

Suggest **two** strategies used in Salt Lake City to effectively manage earthquakes.

You **must** refer to the resource in your answer.

(4)

1 have a plan, you hees to practis what to do if earthquakes heppened

2 When it will heppened keep away from buildings

ResultsPlus Examiner Comments

0 marks. This response does not go beyond direct lifts from the resource which are not awarded credit.

(c) Study Figure 3a in the Resource Booklet.

Suggest **two** strategies used in Salt Lake City to effectively manage earthquakes.

You **must** refer to the resource in your answer.

(4)

1 it prepers you by telling you what to do in a situation of an earthquake.

2 it tells you what to do after the earthquake.

ResultsPlus
Examiner Comments

1 mark. It prepares you on what to do (1) - this is just enough to gain credit as it isn't a direct lift from the resource.

(c) Study Figure 3a in the Resource Booklet.

Suggest **two** strategies used in Salt Lake City to effectively manage earthquakes.

You **must** refer to the resource in your answer.

(4)

1 During preparation, the figure states that Salt lake city educates residents on planning action before the earthquake occurs

2 Salt Lake City also informs residents to set an evacuation strategy, suggesting that they have set evacuation routes to get to high ground.

ResultsPlus
Examiner Comments

2 marks. Educates residents on planning before an earthquake (1) Informs residents to set an evacuation strategy (1)

(c) Study Figure 3a in the Resource Booklet.

Suggest **two** strategies used in Salt Lake City to effectively manage earthquakes.

You **must** refer to the resource in your answer.

(4)

1 Figure 3a suggests keeping/storing resources such as food and water, or possibly first aid equipment, as these limited resources become vital in the scenario of an earthquake

2 Figure 3a suggests staying outdoors in the event of an earthquake, this can effectively reduce casualty numbers as people can avoid collapsing structures or find themselves trapped under rubble with severe injuries

ResultsPlus
Examiner Comments

3 marks. Keep/store resources such as food (1). Stay outside to avoid collapsing structures (1) to avoid being trapped under rubble (1).

(c) Study Figure 3a in the Resource Booklet.

Suggest **two** strategies used in Salt Lake City to effectively manage earthquakes.

You **must** refer to the resource in your answer.

(4)

1 "Prepare and store" Valuable goods like food and water.
↳ This, In the event of the building collapsing and so shortage of resources, a grab bag would allow for survival until emergency services came. Reduces risks of death.

2 Indoors - "drop, cover" and "hold onto something heavy"
↳ would reduce likelihood of being hit by a falling object like a light or concrete, (hiding under tables)
↳ can reduce risk of head injury or body injury.

ResultsPlus
Examiner Comments

4 marks. Store valuable goods like food and water (1) would allow survival until emergency services arrived (1). "Drop, cover hold" would reduce the likelihood of being hit by falling object (1) which reduces the risk of injury (1).

Question 3(d)

Most candidates were able to correctly identify the plate boundary shown in the resource, where this was not the case most candidates had written constructive.

Question 3(e)

This item required candidates to explain two characteristics of volcanoes found at different types of plate boundary. The focus of this question was not on why volcanoes are formed at different plate boundaries, which meant some candidates were not awarded marks as they provided a detailed explanation of why volcanoes are formed at destructive and constructive plate boundaries but no detail about the different types of volcano this results in or why. A very small number of candidates wrote a response about why volcanoes did not occur at transform (conservative) plate boundaries which also was not relevant to the question asked. It is important candidates can explain why shield volcanoes have the characteristics they do and that composite/stratovolcanoes have very different characteristics to shield volcanoes.

(e) Explain **two** characteristics of volcanoes found at different types of plate boundary.

(4)

1 The volcanoes may have different types of material erupted. They can erupt ash, lava, gas ~~and~~ or pyroclastic flow. The different combinations of erupted material, differ from volcano to volcano and plate boundary.

2 Volcanoes found in destructive plate boundary such as Ring of Fire ~~and~~ are extremely dangerous and are very active.

ResultsPlus
Examiner Comments

1 mark. The first point is not specific to a type of volcano or plate boundary and so was not awarded credit. Destructive plate boundary volcanoes are extremely dangerous (1) - this was given the mark for implying they erupt more explosively.

(e) Explain **two** characteristics of volcanoes found at different types of plate boundary.

(4)

1 On destructive plate boundary, the oceanic and continental plates move towards each other. The oceanic plate becomes subducted due to having a higher density. The oceanic plate then melts due to the high heat which builds pressure under the earth's surface and erupts.

2 At a constructive plate boundary, the two plates move apart each other, causing magma to plume out due to friction.

ResultsPlus
Examiner Comments

0 marks. This response gives accurate detail about what happens at each type of plate boundary where volcanoes occur but the focus of the question is on why the volcanoes formed have different characteristics and so this response was not awarded credit.

(e) Explain **two** characteristics of volcanoes found at different types of plate boundary.

(4)

1 In a constructive plate boundary, shield volcanoes form, with sloping sides of cooled lava and ash. The lava is runny and the ~~vola~~ volcano is not very explosive.

2 In a destructive plate boundary, strato-volcanoes form, with steeper sides of layers of cooled lava and ash. The lava is thick and the volcano is very explosive.

ResultsPlus
Examiner Comments

4 marks. Constructive plate boundary has shield volcanoes with sloping sides (1) as the lava is runny (1). Destructive plate boundaries have stratovolcanoes with steeper sides (1) as the lava is thick (1).

Question 3(f)

Candidates were required to explain one reason why people live in areas at risk from volcanic eruptions.

It is often these 3-mark questions which students find most demanding as they often find it challenging developing one point in sufficient depth to be awarded 3-marks. Historically, candidates have performed well on questions focused on reasons why people live near hazards. Where candidates did not achieve full marks this was because they wrote about poverty which was credited but is more challenging to develop sufficiently to gain the third mark as poverty and 'don't have enough money/can't afford to move' are the same idea and cannot be double credited.

(f) Explain **one** reason people live in areas at risk from volcanic eruptions.

(3)

Soil is more fertile, better for agriculture, property is cheaper

ResultsPlus
Examiner Comments

1 mark. Soil is fertile (1) OR property is cheaper (1) - but they are separate points rather than one developed point.

(f) Explain **one** reason people live in areas at risk from volcanic eruptions.

(3)

One reason people live in areas at risk from volcanic eruptions is because the soil around the volcano contains alot of nutrients therefore farmers use that soil to grow crops.

ResultsPlus
Examiner Comments

2 marks. Soil contains lots of nutrients (1) therefore farmers can grow crops (1).

(f) Explain **one** reason people live in areas at risk from volcanic eruptions.

(3)

People who live near active volcanoes are prepared and knowledgeable about the risks of an eruption. They learned to minimalize the threat (by making sure it's easy to evacuate, buildings are high quality, etc). So they many not feel that the volcano is a that serious of a threat and don't move away.

ResultsPlus
Examiner Comments

3 marks. People living there are prepared and knowledgeable about the risks (1) so they do not feel the volcano is a severe threat (1) and so choose not to move away (1).

Question 3(g)

In these 8-mark extended writing questions candidates are required to blend their use of the resource (AO4) with their own knowledge and understanding of the issue presented (AO3), therefore they are not case study questions, rather they require the candidates to apply their geographical understanding to the context shown in the resource.

In this instance candidates were required to study Figure 3c, a map showing the location of the tropical cyclone and the strength of the wind in two different countries and a fact file of physical factors and impacts of each example.

Candidates were able to access Figure 3c and describe the contrast between the two examples. The focus of this question was on the influence of physical factors on vulnerability and some candidates were able to formulate strong arguments for why the impacts were less despite 'worse' physical factors.

The command word 'analyse' needs to be addressed to achieve full marks and many candidates found this challenging as there was often minimal judgement demonstrated in candidate responses as they often remained descriptive.

(g) Study Figure 3c in the Resource Booklet.

Analyse how physical factors can contribute to a country's vulnerability to tropical cyclones (typhoon).

You **must** refer to the resource in your answer.

(8)

Physical factors of typhoons are relevent to the contries vulnerability when the hazard hits.

The death toll in China was 5 deaths while in the Philapeens it was 470 deaths. The category for China was 4 while in the Philapeens it was 5. ~~Even though~~ The category in China was lower, the death count was much less than the philapeens.

The diameter of the storm in China was 926km while in the Philipines it was 833 km. The peak wind speeds in China was 249 km/h while in the Philapines it was 272 km/h. The data is simila here, both

typhoons being very big and destructive. It is no surprise that both countries had landslides and had a similar cost in damage of $1.2 billion (China) and $1.05 billion in the Philapsines. This damage was likely to infrastructure because of the landslides.

Overall, the physical factors of each typhoon effected the physical, and social sides of both countries. They were similar in the built of the typhoon but the category and death toll was very different. This may mean China is more prepared for typhoons

ResultsPlus
Examiner Comments

Level 1 - 3 marks. Basic comparative statements about 3 different factors which use the data shown in the resource to state which country had the higher or lower data. There is an attempt at a conclusion with a simple statement that China (means Japan) was more prepared. There is some evidence of AO3 and stronger evidence of AO4 to reach the top of level 1.

(g) Study Figure 3c in the Resource Booklet.

Analyse how physical factors can contribute to a country's vulnerability to tropical cyclones (typhoon).

You **must** refer to the resource in your answer.

(8)

Figure 3c shows two countries Japan and Phlippines and how a cyclone has affected bg country. Japan is a weee-developed country while Philippines is a developing country so, Japan prediction and prevention methods are said to be better than the philippines one also, ~~the~~ Philippines has been hit by a catogary 5 storm while Japan is hit by a catogary 4 storm and the wind speed of Philippines typhoon is 272 km/h when Japan's is 249 km/hr so damage is more in Philippines compared to Japan but the strom sarges in Japan are 14m whereas Philippines only had 4m. so, more damage has happed in

Japan when considering strom surges. It damage also depends on how long the typhoon lasted and also the population of the country. There is more amount of precipitation in Japan about 500 mm in 24 hrs and 150 mm in Philineen in 24 hrs. How the countries able to copa with the natural hazard are the resuce teams effective is food and water is availabo for everyone and distributed properly by the authorities.

ResultsPlus
Examiner Comments

Level 2 - 4 marks. This response has two ideas discussed with an attempt to link to why this could make one place more vulnerable. There is a basic attempt to analyse by bringing in the idea it is preparation impacting how countries are able to cope. There needs to be greater depth of AO3 to move higher into level 2.

(g) Study Figure 3c in the Resource Booklet.

Analyse how physical factors can contribute to a country's vulnerability to tropical cyclones (typhoon).

You **must** refer to the resource in your answer.

(8)

According to figure 3c there are many factors that can contribute to vulnerability such as wind speed, the category and storm surge. Cyclone Nanmadol was weaker than cyclone Rai by only 1 category. However, it has a larger storm surge height meaning that more water is forced onto land after the cyclone has occured. This means there is more damage to infrastructure and coastal habitats. It also has a higher cost in damage possibly because the location Japan has invested more into its infrastructure compared to the Philipines.

Wind speed is a major factor because the greater the wind speed the more debris and

items ~~then~~ picked up causing more deaths and damage to infrastructure. Phillipines has a windspeed of 272 km/h which could have resulted in the higher death toll ~~as it was due to flooding~~ which could also link to the countries weaker infrastructure. Landslides also contribute to vulnerability especially for people who live in rural areas. ~~They are not landslides happen quickly~~

ResultsPlus
Examiner Comments

Level 2 - 5 marks. This response demonstrates a clear attempt to make a judgement with language such as 'major factor' within the response. There is a discussion of the differences between the two cyclones and an idea as to why Japan suffered more economically and the role of the wind speed in the social impacts on the Philippines. There is evidence from the resource throughout to move the response to the middle of level 2. To reach the top of the level a wider range of factors needs to be discussed.

(g) Study Figure 3c in the Resource Booklet.

Analyse how physical factors can contribute to a country's vulnerability to tropical cyclones (typhoon).

You **must** refer to the resource in your answer.

(8)

In figure 3c it highlights how the Jebi typhoon had a category 4, 926km diameter and peak wind speed of 249 km/h. In figure 3c it highlights how the typhoon had affected most of South Japan with a windspeed of 100 - 129 km/h while the rest of Japan with 60-99 km/h including Tokyo. Due to Japans high development, the government has been able to put in place different prevention and predictive strategies. Education on how to respond to a typhoon is widespread in Japan, so therefore people know how to respond and effectively reduces deaths. Early warning systems are put in place to warn the population beforehand to take shelter or go to places of lower risk such as shelters to successfully reduce injuries and death. Although the severity of the typhoon were quite high and affected the entirety of Japan as shown in figure 3c, the number of deaths have been kept low at 5 shown in figure 3c because of Japans development and lower vulnerability

due to precaution strategies to the impacts. The storm surge height was 14cm and there was 500mm of precipitation in 24 hours as shown in Figure 3c which also highlights the flooding in japan yet the low deaths. In figure 3c it highlights the impacts of 1,410,000 experiencing power cuts

In figure 3c it shows how in the phillipines the typhoon was a category 5, 633km storm diameter and a peak wind speed of 277km/h. In figure 3c it highlights how the ~~Islands~~ islands each in phillipines have been affected with wind speeds of 275-255km/h and 250-224km/h. The Phillipines is a developing country and doesn't have good preventive strategies or predictive strategies, which contributes to the worst effects of the ~~typ~~ typhoon as highlighted in figure 3c with 410 death as an impact due to the low development.

ResultsPlus
Examiner Comments

Level 2 - 6 marks. There is a clear argument which flows through this response whilst using evidence from the resource to support their point of view with three ideas explained. This response more heavily focuses on the influence of economic/social factors which are not shown in the resource and so although they are valid to justify the impacts shown a stronger discussion of the physical factors was required to move into level 3.

(g) Study Figure 3c in the Resource Booklet.

Analyse how physical factors can contribute to a country's vulnerability to tropical cyclones (typhoon).

You **must** refer to the resource in your answer.

(8)

Firstly wind speeds play a large role in vulnerability. Figure 3c shows that the peak windspeed in Japan during the Nonmadol cyclone was 249 km/h and the highest windspeeds occurred around south Honshu and Shikoku (the least populous out of Japan's 4 main islands). This resulted in a CAT 4 storm which, while dangerous, is less dangerous than Typhoon Rai in the philippines - which had peak wind speed of 272 km/h and the highest wind speeds were at the central islands containing population centres like Cebu. This was a cat 5 storm. In tropical cyclones, wind can uproot trees and cars which can hit people or destroy infrastructure. Since Japan is a developed country it likely has better infrastructure than the philippines - an emerging country. so this means Japan may have been less vulnerable to wind speeds only 190,000 experienced power cuts in Japan compared to 3 million in the Philippines, proving that Japan's infrastructure and technology is better, preventing vulnerability to the cyclone.

Storm surge and precipitation are also physical factors that affect a county's vulnerability. In Japan, the storm surge height was 14 metres and there was more precipitation, at 500 mm in 24 hours, according to Figure 3c. This means that more people were at risk of coastal flooding and tsunami's which may explain their higher incurred costs of $1.2 billion. In Japan, the storm was also large in diameter, meaning it had more energy and could affect multiple regions, from Hokkaido to Kyushu, thus increasing vulnerability. Both countries experienced land slides but philippines had a much higher death toll. This may have been because cyclone Rai affected many small islands that are hard to access and sometimes underdeveloped. So more people were at risk of being crushed by rocks due to landslide or falling buildings. Figure 3c shows that the death toll for philippins was 410, compared to only 5 for the Japanese cyclone.

ResultsPlus
Examiner Comments

Level 3 - 7 marks. This response analyses the factors in some detail and interleaves the evidence from the resource to accurately support each point being made. There is a consideration of a range of factors with some depth to each and although there is no conclusion at the end there is a judgement within the response to meet the level 3 descriptors.

(g) Study Figure 3c in the Resource Booklet.

Analyse how physical factors can contribute to a country's vulnerability to tropical cyclones (typhoon).

You **must** refer to the resource in your answer.

(8)

Figure 3c shows the characteristics of tropical cyclones and the seabed impact on two different countries, the Philippines and Japan.

The Philippines has increased vulnerability to tropical cyclones as wind speeds are significantly higher than, reaching ~~225-259~~ 272 km/h, where as Japan's maximum wind speed is ~~100-129 km/h~~ 249 km/h as peak but ~~usually~~ mainly only 100 - 129 km/h. Additionally, figure 3c shows Japan only had a category 4 storm, whilst the philippines had a category 5 storm. ~~than~~ However, Japan's storm diameter was 926 km/h, & the philippines was 833 km/h. This shows the Japanese cyclone typhoon, Nanmadol, was more widespread. Additionally, the figure shows its storm surge height was 14 m, which is 10m higher than the Philippines.

Furthermore, Nanmadol also caused significantly more rain, at 500 m in 24 hours in comparison to 150m in The Philippines. Although the typhoon in Japan came with more adverse effects, which may be due to less coastal surface area to dissipate energy, both typhoons still caused landslides, which is a type of mass movement where large amounts of rock and sediment are lubricated via the precipitation, and thus saturated which increases the mass, causing it to slide at a constant speed due to the mass along with the force of gravity. Additionally, The Philippiens endured worse impacts. The Philippines had 410 deaths, in comparison to Japans' 5, and 3 million experienced power cuts. ~~Alongside that, the~~ However, Japan has more cost of damage by a bit as it is more developed, so more investments lost creates more economic loss. This also suggests

(Total for Question 3 = 25 marks)

TOTAL FOR SECTION A = 50 MARKS

the Philippines didn't protect and prepare well as they had significantly worse impacts due to their low lying land which may have induced flooding, causing power outages.

Overall, Figure 3c shows low lying underveloped land is more vulnerable.

ResultsPlus
Examiner Comments

Level 3 - 8 marks Although, this response is not perfect it has sufficient range and depth of AO3 and frequent use of the resource to support their ideas with an analysis of each factor and a summing up statement at the end of the response to reach the top of level 2.

Question 4(a)

Question 4(b)

Question 4(c)

Question 4(d)

Question 4(e)

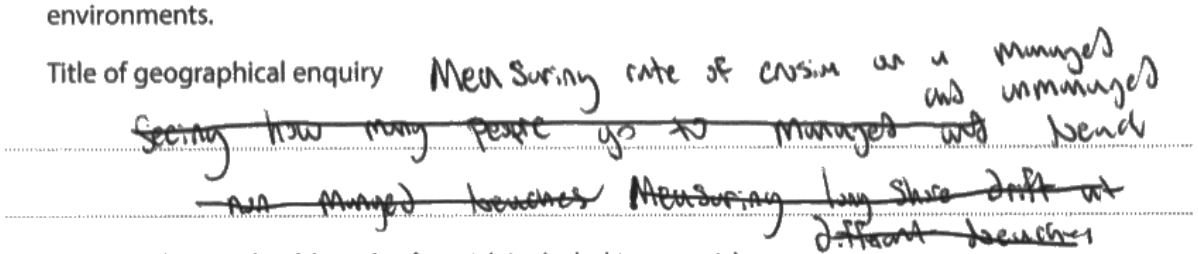
Question 5(a)

Please note that the comments made on Q.5 (which is the most popular fieldwork question on the regional paper) also apply to Q.4 and Q.6 as the questions are in parallel and the resources very similar. Section B contained short answer questions on familiar fieldwork and the extended 8-mark response on unfamiliar fieldwork this series.

For this question candidates needed to describe one health and safety risk included in their risk assessment. Candidates coped well with the demand of this question, with the vast majority being awarded 2 marks. This was a more open question than previous risk questions as risk assessments contain risk and mitigation strategies and so candidates could be awarded 2 marks for identifying a risk and it's consequence or identifying a risk and it's mitigation strategy.

5 Investigating coastal environments

You have carried out a geographical enquiry as part of your work on coastal environments.



(a) Describe **one** health and safety risk included in your risk assessment.

(2)

One safety risk is possible drowning, students should be careful at the beach.

ResultsPlus
Examiner Comments

1 mark. Risk of drowning (1) - the reference to the beach is not specific enough to link to the risk identified, they would need to make reference to the water/sea/waves/tides to get the second mark.

5 Investigating coastal environments

You have carried out a geographical enquiry as part of your work on coastal environments.

Title of geographical enquiry

Investigating types of waves with a specific area - constructive or destructive

(a) Describe **one** health and safety risk included in your risk assessment.

(2)

People ~~to~~ might get their feet cut by seashells or other sediment in the sand of the beach, meaning that they should wear shoes that are closed toe.

ResultsPlus
Examiner Comments

2 marks. Cut feet (1) from seashells (1). This response did go on to provide a mitigation strategy which could also have been awarded a mark if they had not already been awarded full marks.

Question 5(b)

Please note that the comments made on Q.5 also apply to Q.4 and Q.6 as the questions are in parallel and the resources very similar. Section B contained questions on both familiar and unfamiliar contexts this series.

This item required candidates to explain one method they chose for their data collection which helped them answer their enquiry title. Most candidates gained at least 2 marks for this question, as they identified a variable and correctly outlined how they carried out the method. To meet the explain command word candidates needed to go beyond describing the method and give a clear reason why collecting this data would help answer their enquiry title/question/hypothesis. Less candidates were awarded the 3rd mark as they often only described the data collection method.

(b) Explain **one** data collection method used to help answer your enquiry title.

(3)

We used random sampling method to ~~Know~~ reduce bias in the enquiry. This might make our conclusion more accurate. ~~This~~ As we had to represent a large population we used this method.

ResultsPlus
Examiner Comments

0 marks. This response is answering a question focused on sampling strategy and does not explain a data collection method.

(b) Explain **one** data collection method used to help answer your enquiry title. (3)

and ~~clinometer~~ Clinometer

We used a measuring tape to measure the gradient of the ~~beech~~ beach.

ResultsPlus
Examiner Comments

1 mark. Measured beach gradient (1) - there was no credit awarded for just stating the equipment used to do this.

(b) Explain **one** data collection method used to help answer your enquiry title.

(3)

To measure the slope of the beach, we used metre sticks, a clinometer, a large measuring tape. We used systematic sampling to place the metre sticks at equal intervals from the coastline to the end of the beach. We mesured the distance of these intervals with tape measure. We then measured using clinometer the angle at each metre stick all the way up the beach.

ResultsPlus
Examiner Comments

2 marks. Measured the slope of the beach (1) placed the metre sticks at equal intervals to measure the distance and angle along the beach (1) - this response does not provide a reason why this data collection method was used to help answer their enquiry which was required for the third mark.

(b) Explain **one** data collection method used to help answer your enquiry title.

(3)

We measured beach gradient ~~using~~ at different angles along the transect by using ranging poles and a clinometer. Thus, we were able to plot this data on a line graph and visually see how the beach profile changed from the shore to the back of the beach, and showed beach width, which we were able to compare between sites and identify patterns from east to west of the beach, allowing a more informed conclusion.

ResultsPlus
Examiner Comments

3 marks. Measured the beach gradient (1) along a transect using clinometer and ranging poles (1) to compare between sites to see the difference from east to west along the beach (1).

Question 5(c)

Please note that the comments made on Q.5 also apply to Q.4 and Q.6 as the questions are in parallel and the resources very similar. Section B contained questions on both familiar and unfamiliar contexts this series.

Candidates were required to explain why two of their data presentation techniques were accurate. Candidates need to be clear on the difference between why a data presentation technique is accurate and an advantage of the data presentation strategy. The focus of this question was on how they ensured the data presented was accurate. Candidates were awarded a mark for identifying a way they increased accuracy and a second mark for the development of this. Candidates were not awarded credit for simply identifying a data presentation technique.

Fairly frequently candidates wrote responses focusing on how they ensured their data collection was accurate which was not relevant to the question which focused on accuracy of the presentation technique itself.

(c) Explain **two** ways you ensured your data presentation was accurate.

(4)

1 We used IT the data about the beach profile to present it as a scatter graph which was IT generated so it was accurate and quicker.

2 We had ~~taken photographs~~ field sketch of the site and displayed it along with satellite images of the area which makes it easier for us to compare & check for accuracy.

ResultsPlus
Examiner Comments

4 marks. Used IT (1) to generate the scatter graph as it was more accurate (1). Displayed field sketch with satellite image (1) which made it more accurate to compare the two (1).

(c) Explain **two** ways you ensured your data presentation was accurate.

(4)

1 I calculated the percentage, to make sure comparisions between various sets of data is accurate and not biased.

2 Therso I made sure ~~the~~ the scatter graph had no human errors, when drawing the graph. I drew the graph thrice for checking the accuracy.

ResultsPlus
Examiner Comments

2 marks. No credit for the first point as not clearly linked to the accurate drawing of the data presentation. Drew the scatter graph three times (1) to check for human error (1).

(c) Explain **two** ways you ensured your data presentation was accurate.

(4)

1 We used mean values of the amount of erosion for more reliability in the various sites we investigated, that also gave a chance to compare

2 ~~Person~~ We avoided operator errors by taking values and concluding precisely, that led to very few anomalies.

ResultsPlus
Examiner Comments

0 marks. This response focuses on a data analysis technique and a data collection technique and not data presentation meaning it was not awarded credit.

(c) Explain **two** ways you ensured your data presentation was accurate.

(4)

1 We used smart phone apps to generate our graphs. We recorded data using the ~~inpu~~ ~~gave~~ inputs ~~to~~ the apps and later used them for generating graphs. This reduces human errors and changes can be made easily to the graph accuracy. improving.

2 We also used satellite images ~~from~~ to present our data. We used past and present aerial photographs to show the rates of erosion on the coast. This was reliable as we collected these images from national agencies.

ResultsPlus
Examiner Comments

4 marks. Used smart phone apps to generate graphs (1) which reduces the risk of human error (1). Used satellite images alongside past and present aerial photographs to show erosion rates (1) which was reliable as they were collected from national agencies (1).

Question 5(d)

Please note that the comments made on Q.5 also apply to Q.4 and Q.6 as the questions are in parallel and the resources very similar. Section B contained questions on both familiar and unfamiliar contexts this series.

For this question candidates were required to explain one reason why there might be an anomaly in the data they collected. Generally, candidates coped well with the demands of this question and demonstrated a clear understanding of why anomalies occur.

(d) Explain **one** reason why there might be an anomaly in the data you collected.

(3)

Because not every technique we used was perfect, which means that it always going to be difficult as a true anomalies perfect in our data so rather how many times we check.

ResultsPlus
Examiner Comments

0 marks. 'Not perfect' is too vague to be awarded credit as a reason for an anomaly.

(d) Explain **one** reason why there might be an anomaly in the data you collected.

(3)

One reason was we did not use a ruler for measuring sediment size. This meant that those we used our eyes to judge the size. This was inaccurate and increased the chance of human error possibly causing an anomaly

ResultsPlus
Examiner Comments

3 marks. Did not use a ruler to measure sediment size (1) this meant we used our eyes to judge the size (1) which increased the chance of human error (1).

(d) Explain **one** reason why there might be an anomaly in the data you collected.

(3)

There are some error during the measure process such as the degree is not accurate, or the ~~meaner~~ measurement is not very ~~accu~~ accurate.

ResultsPlus
Examiner Comments

1 mark Errors in the measuring process (1)

(d) Explain **one** reason why there might be an anomaly in the data you collected.

(3)

Anomalyies can be seen in date collection as we might not have taken a measurement poopeerly and might have gone wrong at some point due to minor distractions.

ResultsPlus
Examiner Comments

2 marks. Might not have taken a measurement properly (1) due to distractions (1).

Question 5(e)

Please note that the comments made on Q.5 also apply to Q.4 and Q.6 as the questions are in parallel and the resources very similar. Section B contained questions on both familiar and unfamiliar contexts this series.

It is important to remind candidates to write their responses under the correct enquiry context as a proportion of candidates wrote coastal enquiry responses in the river enquiry and hazardous environments enquiry questions. This limited their responses as only generic statements were plausible to the enquiry context being answered to gain credit.

In these 8-mark extended writing questions candidates are required to blend their use of the resource (AO4) with their own knowledge and understanding of the geographical enquiry stage presented (AO3), therefore, they require the candidates to apply their geographical understanding to the context shown in the resource.

In this instance candidates were required to study Figure 5(a & b)/4/6, an annotated map showing the location of their geographical enquiry and annotated images over different time periods of their study area. This secondary data was used for the students to reach a conclusion.

The focus of this question required an evaluation of the extent to which the secondary data supported the students' conclusion. This is the first time this stage of the enquiry process has been assessed as an 8-mark unfamiliar enquiry question. Candidates generally engaged well with the demands of this question with many being able to identify limitations in the secondary data shown and make some level of judgement as to whether the secondary data supported the conclusion. It was the depth and breadth of the response and judgement which distinguished candidates between Level 2 and 3 responses.

Candidates are provided space to provide their geographical enquiry title. They do not receive marks for this, but it provides context for the response that follows.

To gain level 3 credit the command word 'evaluate' needs to be met which requires a concluding statement at the end of a response for full marks. Although, simply having a concluding statement does not automatically move a response into Level 3, there needs to be a range of ideas in sufficient depth to meet the demands of the level descriptors.

(e) Study the data in Figures 5a and 5b in the Resource Booklet. It shows some information about the secondary data the students used to support their enquiry.

Conclusion: **There is very little change in beach profile between calm and stormy years.**

Evaluate the extent to which the secondary data supported the students' conclusion.

You **must** refer to the resource in your answer.

(8)

~~How~~ The students show good understanding and accuracy in their work, and how there is little to no change, However there is

[Screenshot of candidate work] | *images/4GE1_01R_Q05e_1225004_0561_001011460_023.png*

ResultsPlus
Examiner Comments

0 marks. This comment is too vague for credit.

(e) Study the data in Figures 5a and 5b in the Resource Booklet. It shows some information about the secondary data the students used to support their enquiry.

Conclusion: **There is very little change in beach profile between calm and stormy years.**

Evaluate the extent to which the secondary data supported the students' conclusion.

You **must** refer to the resource in your answer.

(8)

Figure 5a shows the students methodology of using photographs over the years. This is a weak piece of data as it may be edited, given with a false date or not show the

[Screenshot of candidate work] | *images/4GE1_01R_Q05e_1179311_0561_000832565_023.png*

ResultsPlus
Examiner Comments

Level 1 - 1 mark. Basic idea linked to the limitation of using photographs.

(e) Study the data in Figures 5a and 5b in the Resource Booklet. It shows some information about the secondary data the students used to support their enquiry.

Conclusion: **There is very little change in beach profile between calm and stormy years.**

Evaluate the extent to which the secondary data supported the students' conclusion.

You **must** refer to the resource in your answer.

(8)

Secondary data as a whole from in Figure 5a and 5b may be unsupportive as it may be outdated. This means that this data has been ~~incorrect~~ corrected years before, enabling heavy change to occur like in Figure 5B shows Dawlish Beach in May 2010. Changes to the beach profile can differ heavily over the years. As a consequence, the data may be unreliable and inaccurate on the present day.

[Screenshot of candidate work] | *images/4GE1_01R_Q05e_1196990_0561_000831128_023.png*

ResultsPlus
Examiner Comments

Level 1 - 2 marks. A basic AO3 and AO4 mark awarded for reference to the resource and reason why the data may be less useful.

(e) Study the data in Figures 5a and 5b in the Resource Booklet. It shows some information about the secondary data the students used to support their enquiry.

Conclusion: **There is very little change in beach profile between calm and stormy years.**

Evaluate the extent to which the secondary data supported the students' conclusion.

You **must** refer to the resource in your answer.

* bias
* inaccurate data

(8)

In figure 5a, it shows a discordant coastline in the USA. In the figure it shows how it goes from hard, hard, soft, soft which ~~shows~~ shows that even the soft rock is easy to errode it hasn't on the Onslow beach. In figure 5b the secondary data supported the students conclusions, as in the photos even after having experienced high tides the sediment hasn't shifted as it is held in place by the beach vegitation and the sand dunes, in may 2009 it shows no difference to may 2010, as the water doesn't affect the beach or constructive wave - mostly build the beach up and

[Screenshot of candidate work] | *images/4GE1_01R_Q05e_1225047_0561_001011455_023.png*

ResultsPlus
Examiner Comments

Level 1 - 3 marks. This response demonstrates some understanding of coastal processes and links this idea to why there has been little change in the images provided to reach the top of level 1.

(e) Study the data in Figures 5a and 5b in the Resource Booklet. It shows some information about the secondary data the students used to support their enquiry.

Conclusion: **There is very little change in beach profile between calm and stormy years.**

Evaluate the extent to which the secondary data supported the students' conclusion.

You **must** refer to the resource in your answer.

(8)

As shown in figure 5a, Onslow beach is located on an area of sand deposits which is soft rock that erodes quickly.

As shown in figure 5b, during the stormy year the middle of the beach had eroded due to the storm surge and hence looked more curved as compared to calm years.

The secondary data is reliable as there is a reference point used to justify that the pictures taken are of the same place. However there is no mention of the time ~~and~~ or date the pictures were taken.

The pictures can a less reliable source of information as compared to primary research which involves taking measurements and pictures yourself.

As shown in figure 5b, during May 2010 after the stormy year the beach faced erosion due to storm waves.

ResultsPlus
Examiner Comments

Level 2 - 4 marks. Three different ideas linked to evidence in the resource to suggest how reliable the data is. There are some advantages and a disadvantage of the secondary data to begin to show more balance. There is a lack of depth to each point which would be required to move higher into level 2.

(e) Study the data in Figures 5a and 5b in the Resource Booklet. It shows some information about the secondary data the students used to support their enquiry.

> Conclusion: **There is very little change in beach profile between calm and stormy years.**

Evaluate the extent to which the secondary data supported the students' conclusion.

You **must** refer to the resource in your answer.

(8)

As shown in Figure 5a and 5b, the beach experienced a lot of erosion caused by storm waves because it was located in an area of sand deposits which suggests it was more prone to erosion as it was soft rock. As shown in Figure 5a and 5b, in May 2010, the year after the storm year, sand deposits have built up again reversing the change the storm waves caused. It suggest lots of longshore drift assisted by constructive waves began rebuilding the beach which supports the students' conclusion. As shown in Figure 5b, the beach looked flatter in 2009 but [redacted] experienced slight changes in the angles [redacted] in 2010 suggesting changes in the gradient which gives proof to the [redacted] erosion that occured in 2009 by the storm waves. Overall, the secondary data supported the students conclusion on a medium scale as no information was provided about what time of day the experiment was conducted and what year the geology map was constructed as

both [redacted] pieces of information would be vital to drawing a conclusion because they may alter the data collected and used.

ResultsPlus
Examiner Comments

Level 2 - 5 marks. This response makes a clear judgement which is supported by their explanation of the coastal processes and links to the geology and evidence from the images. There is a balance between AO3 and AO4 throughout the response. More breadth would be required to move this response to the top of level 2.

(e) Study the data in Figures 5a and 5b in the Resource Booklet. It shows some information about the secondary data the students used to support their enquiry.

Conclusion: **There is very little change in beach profile between calm and stormy years.**

Evaluate the extent to which the secondary data supported the students' conclusion.

You **must** refer to the resource in your answer.

(8)

Secondary data is not reliable as it is not collected by the students themselves. Figure 5b are images of the beach, they can not predict the level of sand by a 2d image. This is not reliable, the students may see changes if they visit the beach. Figure 5b shows increase in vegetation at the beach, this may have affected the ~~wave~~ wave energy and erosion rate. However ~~they~~. the shape of the beach remains almost the same according to 5b. Figure 5a suggests that Onslow beach has mainly sand deposits so has soft rock, these erode easily, suggesting there may be erosion and a change in beach profile, but 5b doesn't show it. Figure 5b does not support the conclusion because the reference point moves closer to the camera, every year, suggest the ~~area~~ area is eroding and the beach is moving back. This is supported by 5a as the rock is soft. Figure 5b also shows vegetation increases in stormy years as there is more water. The students conclusions are not valid or reliable as they were based on secondary data not primary.

The figures don't support the conclusion, as there is a change in reference point and the gradient should be changed too as soft rock would be eroded easily. The gradient is not clearly visible due to the image being 2d and at an angle. No clinometers or ranging poles used so not reliable. The secondary data has no statistics either.

ResultsPlus
Examiner Comments

Level 2 - 6 marks. This response has a broader range of ideas explained which are all supported by accurate evidence from the resource. Each idea contains a simple judgement as to whether the data being discussed supported the candidate's conclusion but there is a lack of overall judgement which is required to move into level 3.

(e) Study the data in Figures 5a and 5b in the Resource Booklet. It shows some information about the secondary data the students used to support their enquiry.

Conclusion: **There is very little change in beach profile between calm and stormy years.**

Evaluate the extent to which the secondary data supported the students' conclusion.

You **must** refer to the resource in your answer.

(8)

One strength of the secondary data is that it is visually appealing and one is able to tell the difference between the sites every year before and after the storm. However, a For example, on 5b, there is significant difference between the sites every year this allows the students to draw the conclusion between whether or not easily spot trends to support their conclusion - However, a weakness of this is that it does not show other parts of the site this means that the students may have drawn an incorrect conclusion, as there is no context in the photograph - the other sites may have been affected more - To improve, the students should collect data from multiple parts of the site to ensure full accuracy of conclusions drawn. The data is therefore largely unreliable, due to limited site view, and a restricted perspective of the beach. Therefore, it supportive it supported the conclusion, the lack of context could have brought about an incorrect as shown in figure 5a,

Another strength of the secondary data is that it shows the types of rock in a visually appealing manner, and it is colour coded for easy viewing this allows them to use the data with

are to support their conclusion. However, the data is not specific to the inquiry question as it does not show any difference between before and after. It is also difficult to find a correlation between rock type and how this affects the beach in stormy or calm years. To improve, instead of looking at the rock type of "loose sediments", the students should consider a source which differentiates before and after. Therefore, the data / conclusion is highly unreliable as it is not specific.

In conclusion, while both 5a and 5b are visually appealing, a lack of context and specificity affects the conclusion drawn. Thus, the sources do not support the conclusion much

(Total for Question 5 = 20 marks)

ResultsPlus
Examiner Comments

Level 3 - 7 marks. This response works through the secondary data shown and considers the strengths and limitations, in places there are improvements suggested. There is a clear evaluation throughout the response and a concluding statement at the end of the response. A more substantiated conclusion would be required to reach the top of level 3.

(e) Study the data in Figures 5a and 5b in the Resource Booklet. It shows some information about the secondary data the students used to support their enquiry.

Conclusion: **There is very little change in beach profile between calm and stormy years.**

Evaluate the extent to which the secondary data supported the students' conclusion.

You **must** refer to the resource in your answer.

(8)

Fig 5a shows the geology map for Onslow beach, North Caroling, USA. It shows the presence of 4 types of rocks - Igneous, metamorphic, loose sediments and sand. Although loose sediments are softer rocks it seems to have lesser erosion in the South. Students using this map and photographs may face errors in their data collection because photographs do not show the changes that have taken place in those areas. They only show us the area at a point in time. Therefore, drawing geology maps as such can also lead to error. We cannot be sure of the accuracy of the map. Fig 5b suggest 3 pictures that are a secondary data. Secondary data helps us collect information which may not be able to be collected through primary data collection methods. It shows pictures of the same part of the beach from the years 2008, 2009 - during the stormy year, and 2010 - during the calm year. It is certainly not posible to derive conclusions on the change in the beach profile and gradient, without using a clinometer (or) any kind of equipment

The picture from May 2010, shows that there was erosion at the back of beach due to storm waves but there is no mention of the length (or) width of the beach in other years. Therefore, we cannot be sure about the occurrence of erosion from the picture. Also, there is no mention of change in the coastline after erosion took place in the geology map. The changes could have been drawn and represented in a bar graph as they are quicker to draw, easy to interpret and can be IT generated. The Key reference shown in the figure cannot always show that it is the same location.

(Total for Question 5 = 20 marks)

In conclusion, the students should have visited the site and used equipment like the clinometer to measure the beach profile, gradient and width using a tape measure. Therefore, this data is not reliable and the conclusion's accuracy is pulled down.

ResultsPlus
Examiner Comments

Level 3 - 8 marks. This response has a clear argument throughout emphasising the importance of needing gradient data to make an accurate conclusion which is lacking throughout the secondary data. Although, this response is not perfect there is sufficient depth and breadth, with evidence of AO3 and AO4 throughout to reach the top of level 3.

Question 6(a)

Question 6(b)

Question 6(c)

Question 6(d)

Question 6(e)

Paper Summary

Areas for focus for next series.

Based on their performance on this paper, candidates are offered the following advice:

- Ensure candidates make specific reference to the resource shown for 4-mark questions in Section A.
- Candidates need to know that 'pollution' on its own will not be credited, the type needs to be qualified.
- Candidates need to be confident in evaluating different stages in the geographical enquiry process not just fieldwork methods.
- Candidates need to be aware of the difference between data presentation and data collection methods.
- Candidates need to be able to explain how they ensured their data presentation was accurate.
- When answering the 8-mark longer response questions candidates need to be clear on the demands of the command word 'analyse'. This requires candidates to investigate an issue by breaking it down into different components and making logical, evidence-based connections between these components. There is an expectation that judgement is made, although, this does not need to be in the form of an evaluation statement at the end of the response.
- Candidates need to recognise that the longer response 8-mark question is dominated by the AO distribution (4 marks for AO3 and 4 marks for AO4). Therefore, responses that are unbalanced or focus on one AO will be limited to a Level 1 response.
- Candidates who just lift text directly from a resource will not gain credit.
- In questions where candidates are asked to develop a single reason, it is important to ensure that the appropriate number of links in the explanatory chain are developed. The number of marks should be used as a guide. These questions usually have the command word, 'suggest' or 'explain', but may differ in depth depending on the expectation of the question.
- Candidates need to ensure they complete the correct fieldwork question; there were a number of candidates answering questions based on a coasts enquiry in both the river and hazard enquiry questions. This limited responses as marks could only be awarded for generic ideas plausible for the enquiry context being answered.
- Some candidates do not follow the instructions on the front of the exam paper and attempt to answer all questions which often results in the candidate running out of time. It would be useful to spend time with candidates to ensure they are familiar with the structure of the paper to avoid rubric infringements as there was an increase in this.

Grade boundaries

Grade boundaries for this, and all other papers, can be found on the website on this link: https://qualifications.pearson.com/en/support/support-topics/results-certification/grade-boundaries.html

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